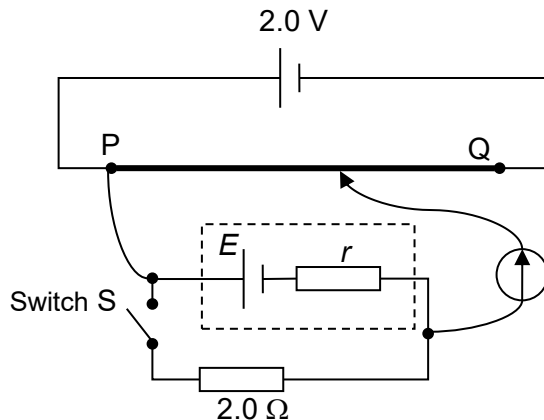


- 20** The diagram below shows a potentiometer circuit used to determine the internal resistance r of a cell of e.m.f. E . The driver cell has an e.m.f. of 2.0 V with negligible internal resistance and the resistance wire PQ is 1.0 m long. The cell is connected in parallel with a resistor of $2.0\ \Omega$. When the switch is open, the balance length is 0.70 m and when the switch is closed, the balance length is 0.50 m .

What is the value of r ?



A $0.15\ \Omega$

B $0.40\ \Omega$

C $0.50\ \Omega$

D $0.80\ \Omega$

Ans: D

When the switch is open, at balance position, there is no current passing through the unknown emf cell.

$$E = \frac{0.70}{1.0} \times 2.0 = 1.4 \text{ V}$$

When switch is closed, at balance position, there is current flowing in the secondary circuit below.

$$E - Ir = \frac{0.50}{1.0} \times 2.0$$

$$1.4 - \left(\frac{1.4}{2.0 + r} \right) r = 1.0$$

$$r = 0.80 \, \Omega$$

21. A charged oil droplet of mass m is falling initially freely in a vacuum between two horizontal